GCE

Rewarding Learning

Chief Examiner's and Principal Moderator's Report

Digital Technology



Summer Series 2023

Foreword

This booklet outlines the performance of candidates in all aspects of this specification for the Summer 2023 series.

CCEA hopes that the Chief Examiner's and/or Principal Moderator's report(s) will be viewed as a helpful and constructive medium to further support teachers and the learning process.

This booklet forms part of the suite of support materials for the specification. Further materials are available from the specification's microsite on our website at www.ccea.org.uk.

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GCE DIGITAL TECHNOLOGY

Chief Examiner's Report

Subject Overview

Direct comparisons of the performance of this session's cohort with those of pre-Covid candidates is not possible due to the difficulty of assessing the lingering, if reduced, difficulties they experienced, as well as taking into account the impact of the special arrangements put in place by CCEA to mitigate their impact. Instead, this report highlights the strengths and weaknesses of the 2023 cohort while avoiding speculation about any lingering impact of Covid on the candidates.

However, it is important to repeat the main general points made in previous reports.

- Candidates must be familiar with DT terminology and must use appropriate DT terminology in their responses.
- Where a question is framed around a context, candidates must address this context as this is an important element of the specification's AO2 (Assessment Objective 2) assessment.
- Questions which are identified as assessing QWC are also a key element of the specification's AO3 assessment. The standard of response required for these questions is much higher, in respect of technical understanding, the accurate use of appropriate DT terminology and the quality of written communication, as is indicated by the unit comments below and by the appropriate part of the mark schemes.

Assessment Unit AS 1 Approaches to System Development

Q1 **(a)** Most candidates could describe two roles of the programmer, but a significant number described variations of producing source code which was excluded by the question stem.

(c) Most candidates could identify three methods. Some candidates carelessly referred to 'documents' instead of 'document sampling' or 'observing' instead of 'observation.'

(d) This question proved to be very discriminating. Few complete answers were noted with many candidates failing to provide valid reasons for their choices. Many simply repeated the requirement.

Q2 **(a)** This question proved to be very discriminating. Many excellent complete or almost complete diagrams were evident. Common errors included the use of incorrect symbols and a failure to identify data stores and dataflows.

(b) Many excellent and complete descriptions were produced, particularly for the use of storyboarding. Candidates frequently used very loose language when describing the use of prototyping such as 'incomplete' or 'get the idea of the UI.'

Q3 **(a)** This QWC question, purposely designed to identify the top candidates, required candidates to evaluate the two changeover methods for a hotel room booking system, in a concise, focused answer using relevant DT terminology and with a high standard of QWC. As expected, very few candidates met the strict criteria for a top-band response.

(b) **(i)** Most candidates could identify four types of user documentation.

(ii) This question proved to be very discriminating. While many candidates could identify two components of technical documentation, few candidates could state clearly how each would be used in corrective maintenance as required.

(iii) Most candidates were familiar with two other types of software maintenance and could describe them as required.

Q4 **(a)** **(i)** Many candidates struggled to explain data conversion clearly and concisely as was required.

(ii) This question proved to be very discriminating. Only a small number of candidates addressed the context of the question – a manual information system being replaced by a computerised system.

(b) **(i)** There was a very high level of response to this question.

(ii) While most candidates were familiar with acceptance testing, some struggled to focus their answer on the specific involvement of end users.

Q5 **(a)** **(i)** Only a very small number of candidates were familiar with the required level with the Agile approach resulting in many imprecise answers which could have described nearly any approach to system development.

(ii) This QWC question, purposely designed to identify the top candidates, required candidates to evaluate the two software development approaches for a system requiring a short development time, in a concise, focused answer using relevant DT terminology and with a high standard of QWC. As expected, very few candidates met the strict criteria for a top-band response.

(b) **(i)** Most candidates could identify 'resources' as was required.

(ii) This question proved to be very discriminating. Most candidates struggled to explain concisely the nature of a constraint or a risk. Many examples were lazy in the extreme such as 'cost,' 'time' or 'problems with ….'

(iii) Most candidates were familiar with the concept of a Gantt chart but many failed to provide the required detail. Many also failed to explain how a project manager would use a Gantt chart.

Q6 **(a)** Many candidates were familiar with an IDE, but a significant number described features related to keying in and running programs which was excluded by the question stem.

While some excellent explanations were evident, many candidates could not explain the terms using the required technical language.

(c) Only a very small number of candidates could understand the algorithm and identify the errors as was required.

(d) This QWC question, purposely designed to identify the top candidates, required candidates to evaluate the object-oriented approach with respect to faster software development, in a concise, focused answer using relevant DT terminology and with a high standard of QWC. As expected, very few candidates met the strict criteria for a top-band response.

Assessment Unit AS 2 Fundamentals of Digital Technology

Q1 **(a)** The overall level of response was very high.

(b) Most candidates stated the correct number of values.

(c) Most candidates produced the correct bit pattern.

(d) **(i)** Most candidates could convert the byte to decimal, but a significant number of candidates omitted the sign of their answer.

(ii) This question proved to be very discriminating with a number of candidates not understanding the phrase 'the most significant bit … flips'.

(iii) Many candidates struggled to explain clearly how we know the sign of the number has changed.

(e) This QWC question, purposely designed to identify the top candidates, required candidates to compare ASCII and Unicode with respect to their representation of characters, in a concise, focused answer using relevant DT terminology and with a high standard of QWC. As expected, very few candidates met the strict criteria for a top-band response.

Q2 **(a)** The overall level of response to this question was very high.

Most candidates could correctly identify three factors.

Q3 **(a)** **(i)** While a small number of candidates confused data verification and data validation, most candidates identified double entry as required. Some descriptions lacked detail, particularly the nature of the checking involved.

(ii) While a small number of candidates confused data verification and data validation, most candidates identified a presence check as required.

(b) For range check, some complete and excellent answers were provided but many responses lacked the required detail and, for example, referred to max and min, or between two values. For lookup, a common error was to describe the use of a drop-down list.

(c) Most candidates could perform the required calculations correctly. Some candidates failed to state explicitly how the error would be detected.

Q4 **(a)** **(i)** Some excellent answers were provided but many candidates failed to provide the required technical detail.

(ii) Most candidates could identify a relevant factor but many struggledprovide the required detail.

(b) Most candidates could explain secondary storage in the required detail.

(c) This question proved to be very discriminating. Some excellent responses were given but many candidates struggled to state how text could be input as was required.

(d) This question proved to be very discriminating. Only a small number of candidates could describe zipping in the required level of detail.

(e) This QWC question, purposely designed to identify the top candidates, required candidates to compare bitmap and JPEG with respect to image quality, in a concise, focused answer using relevant DT terminology and with a high standard of QWC. As expected, very few candidates met the strict criteria for a top-band response.

(f) This question proved to be very discriminating. Some excellent responses were given but many candidates struggled to explain the fundamental difference between backup and archive software.

Q5 **(a)** Most candidates were familiar with batch processing to the required level of detail but many failed to make any specific reference to flight reservations as was required.

(b) Almost all candidates were completely unfamiliar with multithreading, most assuming it was similar to multi-tasking.

(c) This QWC question, purposely designed to identify the top candidates, required candidates to compare two different methods of delivering software using the Internet, in a concise, focused answer using relevant DT terminology and with a high standard of QWC. As expected, very few candidates met the strict criteria for a top-band response.

Q6 **(a)** The majority of candidates were not properly familiar with these two terms in the context of the Web. Most candidates described a server and/or referred to a broadband contract.

(b) Most candidates were broadly familiar with HTML and how it is used to create web pages.

(c) While many candidates were familiar with the term, few candidates could explicitly address the issues of controlling the appearance of pages on a web site.

Assessment Unit A21 Information Systems

Q1 **(a)** Most candidates could identify software, but the other resource was identified by only a small number of candidates. Some candidates identified hardware elements although this was part of the stem, others referred to files which suggested a misunderstanding of the nature of software.

(b) The overall standard of response was very good, and many complete answers were provided. Some candidates ignored the restriction that the same response should not be used more than once.

(c) Most candidates were familiar with the three network resources but many struggled to provide the required detail or addressed the issue of how these were used in a network. Some candidates used very imprecise descriptions such as 'checks the address of data,' 'dissimilar devices' and 'two different media.'

(d) **(i)** The main features of a bus network were familiar to most candidates and complete answers were common. Some candidates focused on the advantages and/or disadvantages of a bus network which was not relevant to the question.

(ii) This QWC question, purposely designed to identify the top candidates, required candidates to compare the two topologies with respect to adding a new node, in a concise, focused answer using relevant DT terminology and with a high standard of QWC. As expected, very few candidates met the strict criteria for a top-band response.

Q2 **(a)** This standard definition of an important DT term caused problems for many candidates in what was intended to be a very accessible question. This highlights the reluctance of many candidates to use DT terminology appropriately without diluting it by paraphrasing.

(b) **(i)** Most candidates could identify the three layers as required.

(ii) Most candidates could identify the OSI set as required.

(iii) Most candidates struggled to provide any relevant technical detail beyond the layers with which these two layers interact. Very imprecise descriptions were common such as 'converts data,' 'light,' 'splits data up' and 'sends data to the right place.'

(iv) Only a small number of candidates could identify any advantage of the OSI model. Very imprecise claims were common such as 'it's easy' or 'it helps users'.

(c) Very few candidates were familiar with the two communication protocols at all, let alone the required detail. The Ethernet protocol was reduced to an Ethernet cable, VoIP equated with 'calls over the Internet.'

Q3 **(a)** The majority of candidates struggled to explain the two terms, few answers provided any of the required detail. Bandwidth was frequently equated with speed, broadband with 'high/good bandwidth.'

(b) **(i)** Those candidates who identified 'fibre optic' could usually provide the required detail.

(ii) This QWC question, purposely designed to identify the top candidates, required candidates to evaluate metal cables and fibre optic cables with respect to data security, in a concise, focused answer using relevant DT terminology and with a high standard of QWC. As expected, very few candidates met the strict criteria for a top-band response.

(c) **(i)** As has frequently been the case in previous sessions, many candidates struggled to explain the use of a parity bit in a concise manner.

(ii) Not all candidates were familiar with CRC. Those that were frequently failed to provide any of the required detail. Most candidates failed to refer to the correction ability of CRC and so could not access full marks.

Q4 **(a)** The overall level of response to this question was very high. Data duplication was better understood than referential integrity and so the question discriminated as intended. A significant number of candidates failed to use the required database technology terminology – answers such as 'data is repeated' and 'stored in the same place.'

(b) Most candidates could explain why the data was not in 1NF. Common mistakes were a failure to refer to the given data or the use of incorrect terminology such as 'repeating data.'

(c) (d) & (e) The overall standard of performing normalisation was very high but candidates must be reminded that they must give the name of each entity, identify the key field(s), and show the remainder of the fields.

(f) The overall standard of the ER model produced by candidates was very high and most candidates used the standard notation as was required.

Q5 **(a)** **(i)** Most candidates accurately described the purpose of the SQL code as creating a table.

(ii) Most candidates accurately described the purpose of the SQL code as inserting a record.

(iii) Most candidates failed to identify that a null value will be inserted into the Tele field.

(iv) Most candidates accurately described the purpose of the SQL code as creating a query.

(b) **(i)** Most candidates could identify correctly the two other components of an expert system.

(ii) Candidates were more familiar with heuristic knowledge than factual knowledge.

(iii) This QWC question, purposely designed to identify the top candidates, required candidates to evaluate the use of an expert system to diagnose illnesses, in a concise, focused answer using relevant DT terminology and with a high standard of QWC. As expected, very few candidates met the strict criteria for a top-band response.

(c) **(i)** In general, candidates were not familiar with pattern matching.

(ii) This question proved to be discriminating as expected. While many candidates could identify one other way of achieving voice recognition, only a few could identify two as required.

Q6 **(a)** In general, the candidature was not properly familiar with cloud computing and its use by businesses. Most candidates answered questions about cloud computing by drawing almost exclusively on their very limited personal experience of storing files 'on the cloud' when using their mobile phones.

(i) Only a very small number of candidates were familiar in any way with hosted instances and clustering.

(ii) Only a very small number of candidates could describe how cloud computing could support data storage and remotely hosted applications in a business context.

(b) **(i)** Most candidates were familiar with the main features of the Computer Misuse Act and some excellent responses were provided. However, many candidates failed to use appropriate language. Phrases such as 'using a computer' or 'accessing data' were common. A small number of candidates confused the Act with the Copyright, Designs and Patents Act.

(ii) While most candidates were broadly familiar with Data Protection Legislation, many failed to address the question which concerned its impact on members of the public.

Principal Moderator's Report

Assessment Unit A2 2 Application Development

Internal Assessment Overview

This is a piece of coursework submitted by Candidates based on a pre-released case study. In this series the case study was BelTel and it carries a 20% weighting of the overall A Level outcomes. Candidates spend 60 hours documenting and producing a relational database solution for BelTel. Centres assess the candidates work using assessment indicators and submit outcomes to CCEA. Moderation of centre outcomes then takes place.

Unit Overview

This was the second year where candidates carried out a problem solving exercise based on the same prereleased case study, BelTel. In the previous series candidates were advised by CCEA to create partial solutions using this case study. This year candidates had to consider a full problem solving exercise , similar to previous series before the pandemic.

The standard of projects submitted was very good and there was a good range of problem solving approaches linked to the case study. Most centres included the samples requested for Moderation in a well organised and well presented manner. Solutions were clearly labelled in electronic folders referenced by Candidate number identifying the top and bottom candidates on their electronic submission. Submission of Password protected databases is not required and can often cause issues during moderation, including access to the database solution.

The majority of centres had teacher comments which proved very useful during the moderation process. Very few centres had no comments at all. The quality of teacher assessment was good and there does seem to be a good understanding of the assessment criteria across centres. Teachers should continue to be encouraged to follow the guidance given in the specification which clearly states that this task is a 60-hour assignment and carries a 20% weighting of the A level award. Centres should also note the reporting back of marking by the moderation team on the TAC 6 report is also useful in preparation for assessing the next cohort. For the most part all centres marked accurately and according to the assessment criteria.

The moderation team felt that centres should put less dependency on encouraging their candidates to follow a predefined template as it leads to very little individual development and doesn't permit candidates to show case their own ability.

The Analysis Section was done well by most candidates. Candidates who were able to clearly identify and document their user requirements produced high quality overall solutions. Candidates who only produced a limited number of requirements found it more difficult producing a detailed design of their solution. There is no need for a fact finding section where the candidates explain the main methods of fact finding. One or two centres continue to do this e.g. interviews, questionnaires, but it is not necessary. There was evidence that the plan or Gantt charts included were limited and did not include much detail. Some basic Gantt charts had no timeframe or no detail in reference to the tasks planned. The inclusion and appropriate use of data flow diagrams (DFD) at level 0 and level 1 has improved across all centres.

In the Design section most candidates explained the different stages of normalisation as well as normalising their data to 3NF. It should be stressed that candidates should explain the stages their data is going through in the normalisation process. Candidates should also include ER diagrams that are linked to the normalised solution. It was pleasing to see well designed data tables with consideration given to data validation. The inclusion of storyboards and wireframes greatly assisted candidates in designing their solutions. Most candidates are showing what algorithms they plan to use in the form of pseudocode or flow charts. Designs of interfaces, forms and reports were good and of a high standard. Candidates should be encouraged to include a detailed test plan linked to user requirements. There were some excellent examples of detailed test plans linked to user requirements. If the user requirements are vague the testing plan tends to lack structure. The better candidates linked their test plans closely to the user requirements stated in the Analysis stage.

In the Application Development stage, the quality of databases created was good and the variety of approaches led to effective development of solutions. It was pleasing to see evidence of excellent relational database solutions produced by many candidates. The majority took the pathway of an ordering system whilst a smaller number approached it via the booking system. In the Implementation strategy the more able candidates referenced a range of changeover methods, user training required and the conversion of legacy data to the new system in their documentation. The majority of candidates were able to verify results of tests planned using the test plan created at an earlier stage. There was also clear evidence of corrective action taken as a result of testing by the better candidates.

In evaluation, candidates did discuss the benefits of their solutions, the limitations and were aware of possible extensions. It was pleasing to see candidates using the non-essential user requirements to come up with the possible extensions to the current system. User guides included by the majority of candidates were of an excellent quality in clearly identified sections.

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